

Assessment 1: Smart Refrigerator Image Classifier

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Kaley Carrubba¹, William Pope¹, and Praya Prawira Turganda¹

¹ University of Newcastle, Callaghan NSW 2308, Australia

Kaley.Carrubba@uon.edu.au

William.Pope1@uon.edu.au

PrayaPrawira.Turganda@uon.edu.au

Abstract. This report details the development of a deep learning-based image classification system for a smart refrigerator application capable of categorising 40 distinct food classes. To address the limitations of the baseline dataset, we constructed an expanded dataset of 10,394 images through multi-source external collection and manual capture in retail environments. Our final model utilises an EfficientNet-B0 architecture with transfer learning and a two-stage fine-tuning strategy. We incorporated heavy data augmentation, including Gaussian blur and random erasing, to ensure robustness against noisy and degraded inputs. Experimental results demonstrate that our model achieves an overall accuracy of 97.90% and a mean class accuracy of 97.85% on the benchmark data. The model's lightweight architecture makes it highly suitable for real-world deployment in resource-constrained embedded systems.

Keywords: Deep Learning, Food Classification, Transfer Learning, EfficientNet-B0, Convolutional Neural Networks, Data Augmentation, Smart Fridge

1 Introduction

The integration of computer vision in household appliances is a key component of modern smart-home ecosystems. A primary application is the smart refrigerator, which utilises internal cameras to automatically recognise food items, assisting with inventory management and grocery tracking. This project addresses the challenge of developing a robust image classifier for 40 diverse food categories, ranging from fruits and vegetables to meats and grocery products.

Developing a reliable classifier for this context involves overcoming several challenges, including visual diversity within classes, potential class imbalance, and the need for robustness against environmental noise and occlusion. Furthermore, the model must be computationally efficient to remain viable for deployment on embedded systems typically found in consumer appliances.

In this report, we describe our methodology for data collection and expansion, our selection of the EfficientNet-B0 architecture, and our iterative training strategy. We provide empirical justification for our hyperparameter choices through sensitivity

analysis and evaluate the final model’s performance and suitability for the intended application.

2 Food Classification Task

The objective of this project is to develop a robust deep learning system for a 40-class food classification task, motivated by the AI vision requirements of modern smart refrigerators. The system must accurately categorise a diverse range of grocery items, including fruits, vegetables, and meats, under varying visual conditions. While FoodTest1 provides a baseline for development, the model is ultimately evaluated on its ability to generalise to a hidden, unseen dataset (FoodTest2). Beyond standard classification accuracy, the task emphasises robustness against realistic visual degradation, such as the noise variants provided in the specification. Finally, the solution must be suitable for practical deployment, balancing predictive performance with the computational and power constraints inherent in embedded consumer appliances.

3 Dataset and Data Collection

The initial dataset provided for this task (FoodTest1) contained labelled images across 40 food classes and served as a baseline for model development. However, preliminary inspection revealed limitations in both size and diversity, with relatively few samples per class and constrained variation in image conditions. As a result, it was insufficient for training a robust deep learning model capable of generalising to unseen data.

To address these limitations, an expanded dataset, referred to as *FoodTest1_expanded*, was constructed. This dataset integrates the original FoodTest1 images with additional samples collected from multiple external sources and manual data collection. The goal was to increase both the number of images per class and the diversity of visual conditions, including variations in lighting, background, perspective, and presentation.

A range of publicly available datasets was used to supplement the original data, including Food-101 [1], the Freiburg Groceries Dataset [2], a fruit recognition dataset [3], a large-scale fish dataset [4], a hierarchical grocery dataset [5], a food dataset for machine learning [6], and the Fish-Vista dataset [7]. Images were selectively extracted and mapped to the corresponding food classes used in this assignment. For example, categories such as oysters, steak, cheese, and salmon were sourced from Food-101, while grocery datasets provided coverage for fruits, vegetables, and packaged items. Additionally, species-level labels from the Fish-Vista dataset were used to match classes such as trout and snapper. A preprocessing script was also implemented to automatically extract and map relevant classes from Food-101 into the expanded dataset structure.

To improve dataset realism and better reflect the intended smart fridge application, a substantial portion of the dataset was collected manually. Over 6,000 images were captured by group members across retail environments, including Woolworths, Coles,

and local fish markets. These images represent realistic contexts such as supermarket shelves, bins, and packaging displays. Care was taken to ensure each image focused on a single food category while incorporating variation in angle, distance, and lighting. This manual collection was essential for introducing real-world complexity not typically present in curated datasets.

Some challenges were encountered during data collection, particularly due to seasonal availability and limited accessibility of certain items (e.g., peaches and specific whole fish). While the target was at least 250 additional images per class prior to augmentation, some classes contain fewer samples.

The final dataset comprises over 10,394 images across 40 classes, combining original, externally sourced, and manually collected data. All external datasets were used in accordance with their respective licences: Food-101 is restricted to non-commercial scientific use, the Freiburg Groceries, fruit recognition, and fish segmentation datasets are licensed under Creative Commons Attribution 4.0, and the remaining datasets (hierarchical grocery, food for machine learning, and Fish-Vista) are distributed under the MIT License. All data was used for academic, non-commercial purposes, with appropriate attribution provided in the references and project documentation (README).

To ensure consistency, all images were stored in a single directory and renamed according to the original FoodTest1 naming convention (*item-Number_itemName_photoNumber*). This preserved compatibility with the provided dataset format while enabling seamless integration of additional images. Further preprocessing steps, including data augmentation and formatting for model training, are described in the following section.

4 Data Preprocessing and Augmentation

Prior to training, all images were standardised to ensure consistency across the combined dataset. Images were loaded as RGB and labelled using the two-digit class prefix in the filename, matching the original FoodTest1 convention to ensure 100% compatibility between original and expanded data.

Images were first resized to a resolution of 256×256 , followed by a random resized crop 224×224 during training, which is the required input size for EfficientNet-B0. This process allows the model to learn from varying object scales and framings. For validation and inference, deterministic preprocessing was applied (resize 256×256 followed by a centre-crop 224×224) to ensure evaluations remain uniform and repeatable. Finally, pixel values were converted to tensors and normalised using ImageNet mean and standard deviation to ensure compatibility with pretrained weights and to stabilise training.

To improve generalisation, a robust augmentation pipeline was applied during training to simulate realistic variations and prevent overfitting. Spatial augmentations, including horizontal and vertical flips and random rotations (1° – 25°), ensure the model remains invariant to the item's orientation and position. Photometric augmentations using colour jitter introduce variations in brightness, contrast, and saturation, which are

critical for handling reflections and the diverse lighting conditions found inside consumer refrigerators.

Additional augmentations were chosen to ensure robustness against the noisy inputs specified in the assignment:

- Gaussian Blur (40% probability): Prepares the model for camera blur or low-frequency sensor noise.
- Random Erasing (25% probability): Simulates missing visual information caused by packaging, overlapping items, or partial occlusion in a crowded fridge.

These strategies directly support the smart-fridge task by preparing the classifier for the imperfect inputs common in household environments, maintaining high classification performance even on degraded data.

5 Model Selection and Transfer Learning Strategy

Our final model was based on EfficientNet-B0, using pretrained weights from ImageNet. This means the model did not start from random values. Instead, it began with a network that had already learned general visual patterns from a large-scale image dataset, which were then adapted to the 40 food classes in this task.

Transfer learning was selected due to the relatively limited size of the dataset compared to the requirements of training a deep neural network from scratch. By leveraging pretrained features such as edges, textures, and shapes, the model was able to converge more efficiently and achieve strong performance with less data. This approach also improves generalisation, particularly when working with diverse and real-world image conditions.

EfficientNet-B0 was chosen because it provides a strong balance between classification performance and computational efficiency [8]. This is particularly important for the smart fridge application, where models must operate under constraints such as limited processing power and the need for fast inference. Compared to larger architectures such as ResNet, EfficientNet-B0 achieves comparable performance with fewer parameters, making it more suitable for deployment in resource-constrained environments. While lightweight models such as MobileNet offer faster inference, they may sacrifice accuracy, which is critical for distinguishing between visually similar food classes. Therefore, EfficientNet-B0 was selected as it offers the most appropriate trade-off between accuracy and efficiency for this task.

6 Model Architecture and Modifications

Most of the original EfficientNet-B0 architecture was retained, particularly the pretrained feature extraction backbone initialised with ImageNet weights. These early layers capture general visual features such as edges, textures, and shapes, which remain relevant for food image classification and reduce the need for extensive feature learning from scratch.

The primary architectural modification was applied to the classification layer. The original ImageNet classifier was replaced with a custom head consisting of a dropout layer followed by a fully connected linear layer mapping to 40 output classes. This modification adapts the network to the specific classification task while maintaining the general feature representations learned during pretraining. The inclusion of dropout ($p = 0.3$) helps reduce overfitting by regularising the model, which is particularly important given the relatively limited dataset size.

A two-stage training strategy was implemented to adapt the architecture effectively. In the first stage, the pretrained backbone was frozen, and only the classifier head was trained, allowing the model to learn task-specific class boundaries without disrupting existing feature representations. In the second stage, the full network was unfrozen and fine-tuned end-to-end, enabling deeper feature adaptation to the food classification task. This approach improves both training stability and final model performance by progressively refining the learned representations.

7 Training Setup and Hyperparameters

The model was trained in two stages: first by training the classifier head, followed by fine-tuning the full network. This staged approach improves training stability and allows the pretrained model to gradually adapt to the target dataset. As shown in Fig. 1, accuracy improves rapidly during early epochs and increases further after the transition to fine-tuning, eventually stabilising near 99% with minimal gap between training and validation accuracy, indicating strong generalisation and no significant overfitting.

For the loss function, cross-entropy loss with label smoothing of 0.1 was used. Cross-entropy is well-suited for multi-class classification tasks, as it compares predicted class probabilities with ground truth labels. Label smoothing was incorporated to reduce model overconfidence and improve generalisation.

Optimisation was performed using the AdamW optimiser, which is effective for fine-tuning pretrained neural networks. In the first training phase, a learning rate of 1×10^{-3} was applied to the classifier head. In the second phase, differential learning rates were used: 1×10^{-4} for the backbone and 5×10^{-4} for the classifier head. This allows the pretrained layers to adapt more conservatively while enabling faster learning in the newly added classifier.

The model was trained for 10 epochs in phase 1 and 40 epochs in phase 2, with a batch size of 16 and weight decay of 1×10^{-4} . A cosine annealing learning rate schedule was applied in both stages to gradually decrease the learning rate during training, improving convergence.

To monitor performance and reduce overfitting, a 20% stratified validation split was used. This ensures that each class is proportionally represented in both training and validation sets. For classes with very few samples, the split was adjusted to avoid removing too many training examples. The best-performing model was selected based on validation accuracy and saved for final evaluation.

Hyperparameters were selected through iterative experimentation rather than exhaustive search. Initial values were based on standard transfer learning practices and

refined using validation performance to achieve stable convergence and improved robustness to noisy and real-world inputs.

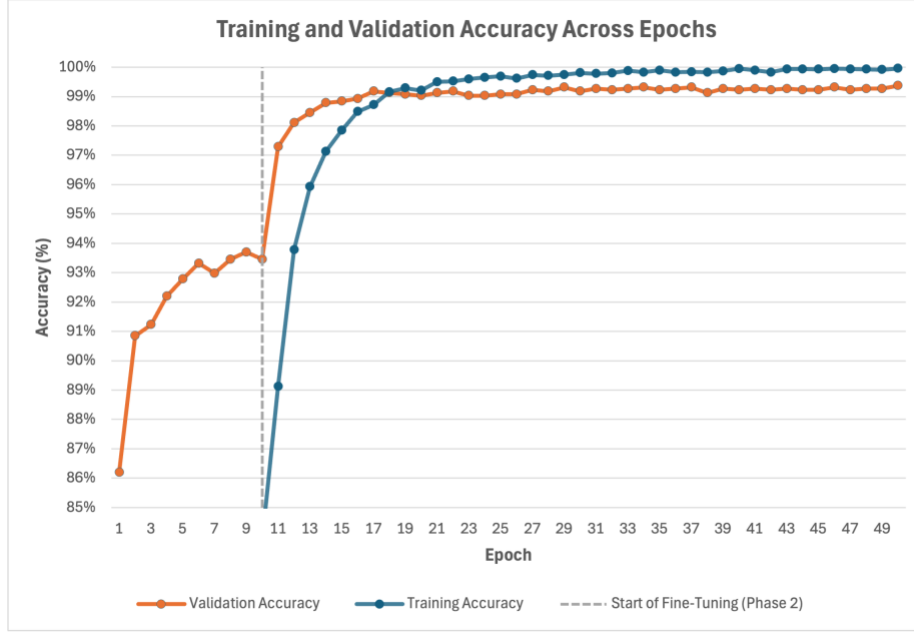


Fig. 1. Training and validation accuracy across epochs for the final model. The dashed line indicates the transition from classifier training to full network fine-tuning.

8 Experimental Variants and Hyperparameter Tuning

To justify the final model configuration, we conducted a sensitivity analysis on a representative subset of the expanded dataset. We tested three variants of the classifier head's learning rate (LR) over a short duration to observe convergence behaviour and stability.

Table 1. Hyperparameter sensitivity analysis results on a 10% data subset.

Variant	Head LR	Best Val Acc (3 Epochs)	Observation
LR_High	1×10^{-2}	76.92%	High initial gains, but became unstable by epoch 3.
LR_Baseline	5×10^{-4}	39.74%	Consistent, stable improvement across all epochs.
LR_Low	1×10^{-4}	10.26%	Learning rate insufficient for timely convergence.

As shown in Table 1, while the high learning rate achieved the highest early accuracy, it exhibited instability with a significant drop in validation performance in the third epoch. Consequently, a more conservative learning rate was selected for the final model to ensure stable convergence during the full training workflow.

9 Final Model and Evaluation

The final model selected for this task was the EfficientNet-B0 architecture, utilising the optimised hyperparameters identified in the sensitivity analysis and trained on the FoodTest1_expanded dataset. This configuration was chosen to balance discriminative power with the computational efficiency required for smart-fridge deployment. The expanded dataset of 10,394 images was divided using a stratified split, allocating 8,329 images for training and 2,065 for validation to ensure the model was exposed to maximum class diversity while maintaining a robust evaluation set.

To evaluate model performance, the trained model was tested using the provided FoodTest1 dataset via the standalone inference script. As summarised in Table 2, the model trained on the expanded dataset achieved substantially higher performance, improving overall accuracy from 93.01% to 97.90% and mean class accuracy from 89.58% to 97.85%.

Table 2. Comparison of model performance when trained on the original dataset versus the expanded dataset.

Metric	FoodTest1 Only	Expanded
Overall Accuracy	93.01%	97.90%
Mean Class Accuracy	89.58%	97.85%

This improvement demonstrates the effectiveness of the data expansion strategy. The additional images enabled the model to learn more robust and generalisable feature representations. Classes that previously performed poorly showed significant gains; for example, cauliflower and peach improved from 0% accuracy to 100%, while steak improved from 66.67% to 100%.

Per-class accuracy results for the final model are shown in Fig. 2. The model achieves 100% accuracy for the majority of classes, indicating strong discriminative capability across a wide range of food categories. However, a small number of classes exhibit lower accuracy, including eggs (50%), sausages (75%), and cheese (88.89%). These cases are likely due to limited sample sizes and visual similarity between certain food items, which can make classification more challenging.

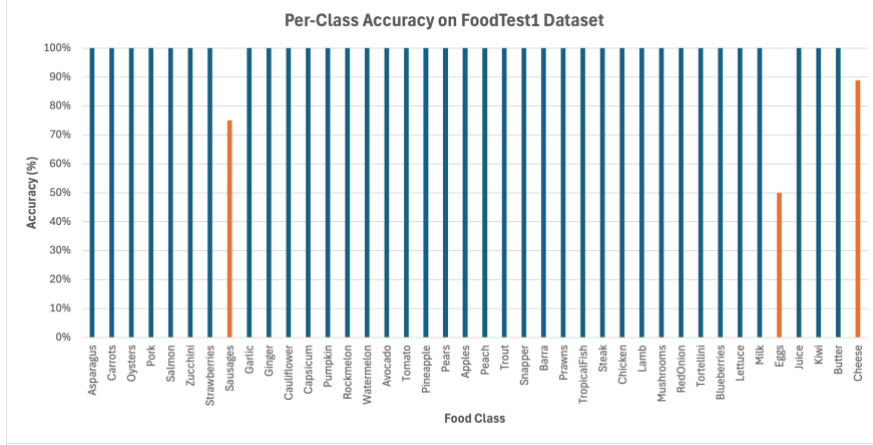


Fig. 2. Per-class accuracy of the final model on the FoodTest1 dataset.

The confusion matrix in Fig. 3 shows that most predictions lie along the diagonal, indicating correct classification across the majority of classes. The normalised matrix highlights that errors are limited to a small number of categories, with the most notable being eggs (50% accuracy) and sausages (75% accuracy). These misclassifications are consistent with the per-class accuracy results and are likely due to limited sample sizes and visual similarity between certain food items.

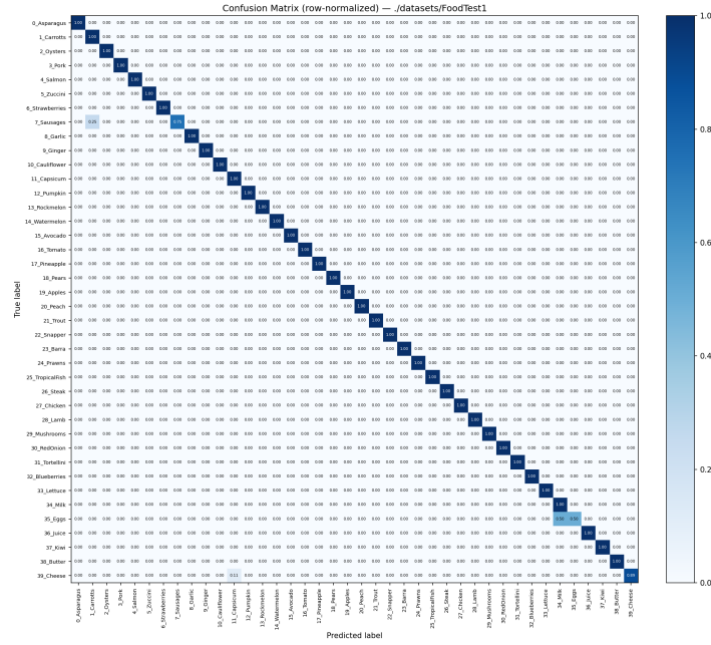


Fig. 3. Normalised confusion matrix of the final model on the FoodTest1 dataset.

Overall, the results indicate that the final model generalises well and performs reliably across most classes. The high overall and mean class accuracy, combined with strong performance on previously difficult categories, demonstrate that the chosen model and data strategy are effective. These findings also confirm that expanding the dataset was a critical factor in improving model robustness and classification performance.

10 Ethical and Deployment Considerations

10.1 Deployment Considerations

For a smart fridge application, the deployed model must balance predictive performance with computational efficiency. In this work, EfficientNet-B0 was selected as the final model due to its favourable trade-off between accuracy, model size, and inference speed. Compared to larger architectures, EfficientNet-B0 has a relatively small number of parameters, reducing both memory usage and computational cost, while still maintaining strong classification performance.

These properties make the model suitable for deployment in resource-constrained environments such as embedded systems. In a real-world smart fridge, inference would need to be performed frequently and potentially on device, where processing power and energy availability are limited. A lightweight model reduces latency and power consumption, enabling faster predictions and more efficient continuous operation.

In addition, the model was designed to generalise to varied and noisy image conditions, which is critical for practical deployment. Unlike controlled datasets, real-world inputs may include occlusion, packaging, clutter, and lighting variation. The use of diverse training data and augmentation strategies supports more robust performance under these conditions. Overall, the selected model provides a practical balance between efficiency and accuracy, making it appropriate for real-world smart fridge use.

10.2 Ethical Considerations

Several ethical considerations arise from the development and use of the dataset and model. Firstly, all external datasets used in this project were sourced from publicly available repositories and utilised in accordance with their respective licences, with all usage restricted to academic, non-commercial purposes. Proper attribution has been provided in both the report and accompanying documentation.

However, it is important to note that some of these datasets, such as Food-101, are distributed under licences that restrict use to non-commercial or research contexts. As a result, while these datasets are appropriate for academic experimentation, they would not be suitable for training a model intended for commercial deployment without additional permissions or alternative data sources. A production-ready system would therefore require either fully licensed commercial datasets or data collected under appropriate usage agreements.

Bias in the dataset is another important consideration. Many publicly available datasets contain images captured in clean or controlled environments, which may not accurately represent real-world conditions. This can lead to models that perform well

on curated data but struggle in practical settings. To mitigate this, a significant portion of the dataset was manually collected in real retail environments, introducing more realistic variation. However, some imbalance and bias may remain, particularly for classes with fewer available samples.

Privacy considerations were also considered during manual data collection. Images were captured in public retail environments with a focus on food items only, and care was taken to avoid including identifiable individuals or sensitive information. This reduces potential privacy risks associated with the dataset.

Despite these efforts, limitations remain. Certain classes are underrepresented due to availability constraints, and the dataset may not fully capture all real-world variations. Future work could involve collecting more balanced data across all classes, expanding coverage of difficult categories, and further improving robustness to diverse environmental conditions.

11 Conclusion

In this project, we successfully developed a high-performance deep learning classifier for a 40-class food recognition task. By expanding the initial dataset to over 10,000 images, we significantly improved the model's ability to generalise to diverse and real-world visual conditions. Our selection of the EfficientNet-B0 architecture, combined with transfer learning and targeted data augmentation, resulted in an overall accuracy of 97.90% and a mean class accuracy of 97.85%.

Our experimental results and sensitivity analysis confirmed that a conservative learning rate and staged training approach were critical for achieving stable convergence. Furthermore, the model's small parameter footprint ensures it is well-suited for the power and memory constraints of smart fridge deployment. Future work could focus on further diversifying the dataset to address remaining misclassifications in visually similar categories and exploring even more lightweight architectures for low-power operation.

12 Appendix

12.1 Implementation Resources

The full implementation used in this project is provided as a standalone directory. The following scripts and files constitute the complete pipeline for data preparation, model training, and inference evaluation.

- “inference.py” – The primary standalone script used to load the trained model and evaluate performance on a dataset folder in the required format.
- “train.py” – The main training workflow implementing the two-phase transfer learning strategy and data augmentation pipeline.
- “sensitivity_analysis.py” – An experimental script used to conduct the hyperparameter sensitivity tests on a 10% data subset to justify final model configurations.

- “run.py” – A unified entry point providing a command-line interface for executing both training and inference tasks.
- “download_data.py” – A utility script that reorganises flat dataset folders into structured, class-specific subdirectories compatible for training.
- “image_renamer.py” – A script used during data collection to standardise file names from external sources into the *XX_ClassName_XXX* convention
- “requirements.txt” – A specification of necessary Python packages and versions, including *torch*, *torchvision*, and *scikit-learn*.
- “README.md” – Detailed documentation regarding project setup, hardware requirements, and execution instructions.

12.2 Full Expanded Dataset

Due to the significant file size (approx. 13 GB), the full expanded training dataset comprising 10,394 images across 40 classes is provided as supplemental material via the following secure cloud storage link:

<https://drive.google.com/file/d/1MnwmRtub7shAAAtLzODawh5jGRVberhcu/view?usp=sharing>

This dataset includes the original FoodTest1 baseline, selectively filtered samples from seven external public datasets, and over 6,000 manually captured images.

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